

This assignment will not be graded and is only for practice.

Key Points: 1

Definition of a vector space V:

A vector space V over $\mathbb{R}$ is a set along with two functions:

$+: V \times V \rightarrow V$ (V is closed under vector addition i.e., if $x, y \in V \Rightarrow x + y \in V, \forall x, y \in V$)

$\cdot: \mathbb{R} \times V \rightarrow V$ (V is closed under scalar multiplication i.e., if $x \in V$ and $c \in \mathbb{R} \Rightarrow c \cdot x \in V, \forall x \in V$, and $\forall c \in \mathbb{R}$)

Axioms of vector additions of a vector space

$$v_1 + v_2 = v_2 + v_1, \forall v_1, v_2 \in V.$$

$$v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3, \forall v_1, v_2 \text{ and } v_3 \in V.$$

There exists an element 0 (called the zero vector of V) in V such that $0 + v = v, \forall v \in V$.

There exists a vector v' in V such that $v' + v = v + v' = 0, \forall v \in V$ (v' is called the inverse (or negative) of v with respect to addition).

Axioms relating vector addition and scalar multiplication of a vector space (since this is usual multiplication in real numbers, we suppress the " $\cdot$ " sign).

For each vector $v \in V, 1v = v$.

For each vector $v \in V$ and for each pair $a, b \in \mathbb{R}, (a + b)v = av + bv$.

For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V, a(v_1 + v_2) = av_1 + av_2$.

For each vector $v \in V$ and for each pair $a, b \in \mathbb{R}, (ab)v = a(bv)$.

Consider a set $V = \{(1, -y) \mid y \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the usual addition and scalar multiplication as in $\mathbb{R}^2$.

Answer the following questions (1) and (2).

1) Which of the following option(s) is (are) true?

1 point

[Hint: Check the axioms of vector space.]

☐ $(1, 2) \in V$.

☐ $(2, 3) \in V$.

☐ $V = \mathbb{R}^2$.

☐ $v_1 = (1, -5)$ and $v_2 = (1, 7) \in V \Rightarrow v_1 + v_2 \in V$.

☐ $v = (1, 9) \in V, c \in \mathbb{R} \Rightarrow cv \in V, \forall c \in \mathbb{R}$.

☐ For all $v \in V, cv \in V$ only for $c = 1$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2) \in V$.

For all $v \in V$, $cv \in V$ only for $c = 1$.

2) Which of the following option(s) is (are) true?

1 point

- ☐ Zero element does not exist in V .
- ☐ Zero element of V is $(0,0)$.
- ☐ Inverse element of $(1, 5)$ is $(-1, -5)$.
- ☐ Inverse element of $(1, 5)$ is $(1, -5)$.
- ☐ Inverse element of $(1, 5)$ does not exist.
- ☐ Inverse element of any element of V does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Zero element does not exist in V .

Inverse element of $(1, 5)$ does not exist.

Inverse element of any element of V does not exist.

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^3$ with the usual addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$, where $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ and scalar multiplication: $c(x, y, z) = (cx, cy, cz)$, $c \in \mathbb{R}$ and $(x, y, z) \in V$. Answer the following questions (3) and (4).

3) Which of the following option(s) is (are) true?

1 point

[Hint: For a vector $v = (x, y, z)$, check whether the condition $x + y + z = 0$ is satisfied or not.]

☐

$$(1, 2, -3) \in V.$$

☐ $(-1, 2, 1) \in V.$

☐ $V = \mathbb{R}^3.$

☐ $(5, -7, 2) \in V.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(1, 2, -3) \in V.$$

$$(5, -7, 2) \in V.$$

4) Inverse element of $(-2, -4, 6)$ with respect to vector addition is

1 point

☐ $(2, 4, -6).$

☐ $(2, -4, -6).$

☐ $(2, 4, 6).$

☐ $(-2, 4, -6).$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(2, 4, -6).$$

5) Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^3$ with the usual addition as **1 point** in $\mathbb{R}^3$ and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}, cz) & c \neq 0 \end{cases} \quad (x, y, z) \in V, c \in \mathbb{R}$$

Which of the following option(s) is (are) true?

[Hint: By adding any two elements in V , or by multiplying a scalar with any element in V , check whether the resultant element is in V or not.]

- ☐ V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
- ☐ V is closed under scalar multiplication i.e., $(c \in \mathbb{R}, v \in V \implies cv \in V)$.
- ☐ V has zero element with respect to addition.
- ☐ $1.v = v$, where $1 \in \mathbb{R}$ and $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.

V has zero element with respect to addition.

6) Consider usual addition and scalar multiplication on $\mathbb{R}^2$ and the subset $V = \{(1, -y) \mid y \in \mathbb{R}\} \subseteq \mathbb{R}^2$. **1 point**

Which of the following option(s) is (are) true?

[Hint: Take two arbitrary elements $(1, -y_1)$ and $(1, -y_2)$ from V and check the conditions given in the options.]

- ☐ For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(a + b)v = av + bv$.
- ☐ For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.
- ☐ For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(ab)v = a(bv)$.
- ☐ The set V with the usual addition and scalar multiplication in $\mathbb{R}^2$ is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(a + b)v = av + bv$.

For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.

For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(ab)v = a(bv)$.

7) Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^3$ with the usual addition as **1 point** in $\mathbb{R}^3$ and scalar multiplication is defined as

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}, cz) & c \neq 0 \end{cases} \quad (x, y, z) \in V, c \in \mathbb{R}$$

Which of the following option(s) is (are) true?

- ☐ For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(a + b)v = av + bv$
- ☐ For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$
- ☐ For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(ab)v = a(bv)$
- ☐ V is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$

Key Points: 2

Definition of a subspace: A subset W of a vector space V is a subspace of V if it is a vector space with respect to the same operations defined on V .

Criteria to check a subset W of a vector space V is a subspace:

Zero vector should be in W i.e., $0 \in W$.

If two vectors v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.

Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$

8) Let $W = \{(x, y) \mid x + 3y = 1, x, y \in \mathbb{R}\} \subseteq \mathbb{R}^2$. Which of the following options are true? **1 point**

- ☐ Zero element exists in W .
- ☐ If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.
- ☐ Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$
- ☐ W is not a subspace of $\mathbb{R}^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.
 W is not a subspace of $\mathbb{R}^2$.

9) Let $W = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a diagonal matrix}\}$ be a subset of a vector space $M_{2 \times 2}(\mathbb{R})$. Which of the following option(s) is (are) true? **1 point**

- ☐ Zero vector does not exist in W .
- ☐ If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
- ☐ Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$
- ☐ W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
 Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$
 W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Key Points: 3

Linear combination: Let V be a vector space and $v_1, v_2, \dots, v_k \in V$. The linear combination of $v_1, v_2, \dots, v_k$ with coefficients $a_1, a_2, \dots, a_k \in \mathbb{R}$ is the vector $v = \sum_{i=1}^k a_i v_i$.

A vector $v \in V$ is said to be a linear combination of $v_1, v_2, \dots, v_k \in V$ if there exist $a_1, a_2, \dots, a_k \in \mathbb{R}$ so that $v = \sum_{i=1}^k a_i v_i$

10) Which of the following vectors are linear combinations of $(2, 1)$ and $(6, 3)$ with usual addition **1 point** and scalar multiplication in $\mathbb{R}^2$?

- ☐ $(2, 1) + 5(6, 3)$
- ☐ $(2, 1)$

☐ (6, 3)

☐ (3, 6)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(2, 1) + 5(6, 3)

(2, 1)

(6, 3)

11) Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ i.e., $v_1 = av_2 + bv_3 + cv_4$, then which of the following options are true? **1 point**

☐ If $a = 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.

☐ If $a = 0$ and $b = 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.

☐ v_2 is a linear combination of v_2 .

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $a = 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.

If $a = 0$ and $b = 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.

v_2 is a linear combination of v_2 .

Your score is: 0/11